Functions: Mini-Lessons

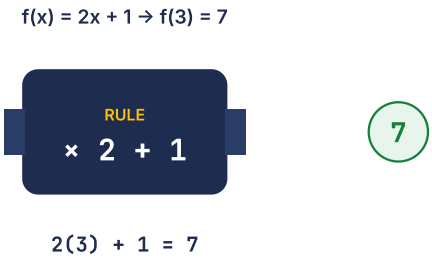
6.1 What is a function? NC.8.F.1

A **function** is a rule where **each input (x) has exactly one output (y)**. f(x) means "the output when the input is x." To find f(4), put 4 in for x.

Input (x)
the number you put in

Output (y or f(x))
the number that comes out

Function
every input has exactly one output



Not a function: one input with **two** outputs, like x = 2 going to both 5 and 7. On a graph, a vertical line that hits the graph twice means it is **not** a function.

Watch out: Two inputs with the same output is still a function. The problem is one input with two outputs.

Walk through: `f(x) = 3x - 2. Find f(4).`

1. What does f(4) ask?
   Put 4 in for x → f(4) = 3(4) − 2
2. Multiply first: 3(4) = ?
   12 → = 12 − 2
3. Finish: 12 − 2 = ?
   10 → f(4) = 10

Check your understanding

1 Score 3 NC.8.F.1

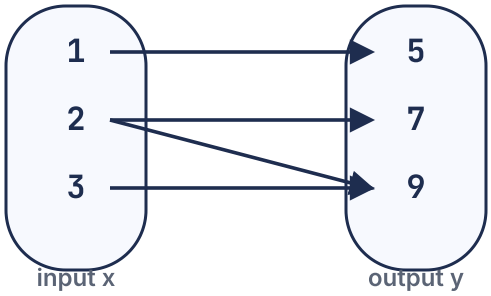
`f(x) = 2x + 5`. Find `f(3)`.

Answer: **11**

1. `f(3) = 2(3) + 5`
2. `= 6 + 5 = 11`

2 Score 3 NC.8.F.1

Is this mapping a function?



A **No: the input 2 has two outputs**

Ⓑ Yes: every input has an arrow

Ⓒ Yes: two inputs share an output

6.2 Slope: rise over run NC.8.F.4

Slope tells how steep a line is. **`slope = rise ÷ run = (change in y) ÷ (change in x)`**. Uphill (left to right) is **positive**, downhill is **negative**, flat is **0**.

Rise
the up/down change: subtract the y-values

Run
the across change: subtract the x-values

Slope (m)
rise ÷ run



`slope = rise ÷ run`

`= (8 - 2) ÷ (3 - 0)`

`= 6 ÷ 3 = 2`

up 2 for every 1 across

Green: positive (uphill) · Red: negative (downhill) · Gray: zero (flat)

Watch out: Rise goes on **top**. Run goes on the bottom. And subtract in the same order for both.

Walk through: `Find the slope from (1, 3) to (4, 12).`

1. Find the rise (change in y).
   12 − 3 = 9 → rise = 9
2. Find the run (change in x).
   4 − 1 = 3 → run = 3

3 Slope = rise ÷ run = ?

9 ÷ 3 = 3 → slope = 3

Check your understanding

1 Score 3 NC.8.F.4

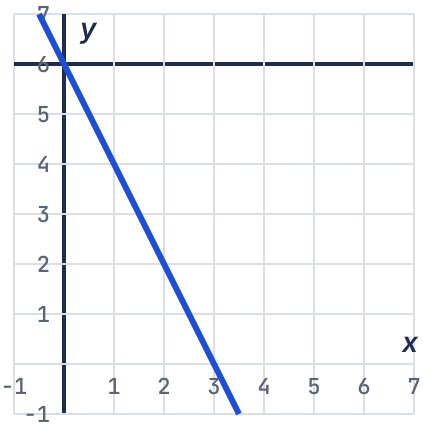
Find the slope of the line through **(0, 2)** and **(3, 8)**.

Answer: 2

1. `rise = 8 - 2 = 6`
2. `run = 3 - 0 = 3`
3. `slope = 6 ÷ 3 = 2`

2 Score 3 NC.8.F.4

What kind of slope does this line have?



(A) **Negative** (B) Positive

(C) Zero

6.3 y = mx + b from a table or graph NC.8.F.4

A line can be written **y = mx + b**. **m** is the slope (how much y changes when x goes up 1). **b** is the y-intercept (the y-value when x = 0, where the line crosses the y-axis).

Slope-intercept form

y = mx + b

y-intercept (b)

where the line crosses the y-axis; the y-value when x = 0

x	0	1	2	3
y	5	8	11	14

y goes up 3 each time → m = 3. When x = 0, y = 5 → b = 5. So y = 3x + 5.

Watch out: In y = 4x − 1, the slope is 4 and the y-intercept is **−1** (keep the sign).

x	0	1	2	3
y	2	5	8	11

+3 +3 +3

`m = 3`

`b = 2`

`y = 3x + 2`

m = slope (rate) · b = y-intercept (start)

Walk through: `Write the equation for this table.`

1

x	0	1	2	3
y	1	3	5	7

How much does y change when x goes up 1?

+2, so m = 2 → m = 2

2 What is y when x = 0?

1, so b = 1 → b = 1

3 Write y = mx + b.

y = 2x + 1 → y = 2x + 1

Check your understanding

1 Score 3 NC.8.F.3

In **y = 4x - 1**, what is the **slope**?

Answer: **4**

2 Score 3 NC.8.F.4

Which equation matches the table?

x	0	1	2	3
y	4	6	8	10

(A) **y = 2x + 4**

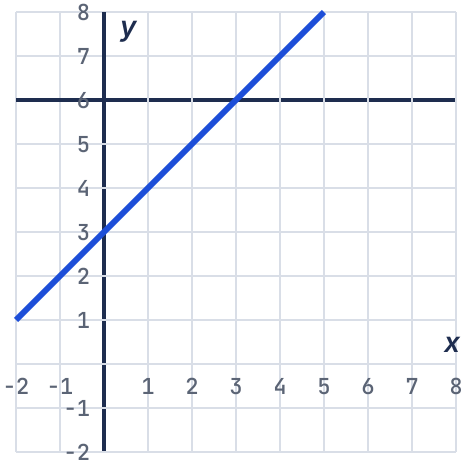
(B) y = 2x

(C) y = 4x + 2

(D) y = x + 4

3 Score 3 NC.8.F.4

What is the y-intercept of this line?



Answer: **3**

6.4 Rate and start value NC.8.F.2

In a real-world line, the **slope is the rate** (per hour, per mile, each day) and the **y-intercept is the start value** (the amount when x = 0, like a fee).

Rate of change
how much y changes for each 1 of x

Initial (start) value
the amount at the start, when x = 0

Increasing / decreasing
a graph going up / going down from left to right



`Taxi: y = 2x + 3`

2 = rate: $2 per mile

3 = start: $3 before driving

`5 miles: 2(5) + 3 = $13`

Phone plan: **`y = 15x + 40`**. $40 to start, then $15 each month.

Watch out: The start value is the number **by itself**, not the number next to x.

Walk through: `A gym charges y = 20x + 50 for x months. What does 50 mean?`

1. In y = 20x + 50, which number is alone (not next to x)?
   50 → b = 50
2. What does the number alone mean?
   The start fee: $50 before any months → $50 is the start (sign-up) fee
3. And 20?
   $20 per month (the rate) → $20 per month is the rate

Check your understanding

1 Score 3 NC.8.F.2

A plant is **`h = 2w + 6`** cm tall after w weeks. How tall was it at the start?

Answer: **6cm**

2 Score 3 NC.8.F.2

A phone plan costs **`y = 15x + 40`**. What is the cost **per month**?

Answer: **15dollars**

3 Score 3 NC.8.F.5

A graph of water in a tank goes **down** from left to right. What is happening?

Ⓐ **The water is decreasing**

Ⓑ The water is increasing

Ⓒ The water stays the same

6.5 Systems: where two lines cross NC.8.EE.8

The **solution of a system** is the point **(x, y) where the two lines cross**. It makes **both** equations true.

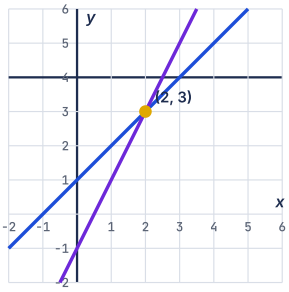
System of equations

two equations used together

Solution

the point that is on both lines

Watch out: A point on only one line is not the solution. Check it in **both** equations.



$y = x + 1$

$y = 2x - 1$

Check (2, 3):

$3 = 2 + 1$ ✓

$3 = 2(2) - 1$ ✓

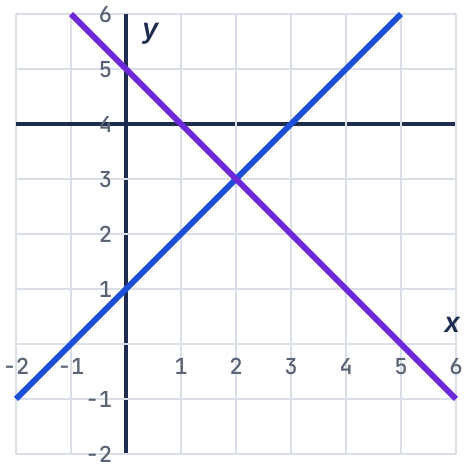
Walk through: Is (1, 3) the solution of y = x + 2 and y = 3x?

1. Check the first equation: y = x + 2 with x = 1.
   1 + 2 = 3 ✓ (y is 3) → y = x + 2: 3 = 1 + 2 ✓
2. Check the second equation: y = 3x with x = 1.
   3(1) = 3 ✓ → y = 3x: 3 = 3(1) ✓
3. Is (1, 3) the solution?
   Yes: it works in both → (1, 3) is the solution

Check your understanding

1 Score 3 NC.8.EE.8

What is the solution of the system shown?



Ⓐ (0, 1)

Ⓑ **(2, 3)**

Ⓒ (3, 2)

Ⓓ (0, 5)

2 Score 3 NC.8.EE.8

Is **(2, 5)** the solution of **y = x + 3** and **y = 2x**?

Ⓐ **No: it works in y = x + 3, but 2(2) = 4, not 5**

Ⓑ Yes: 2 + 3 = 5

Ⓒ Yes: it works in both